

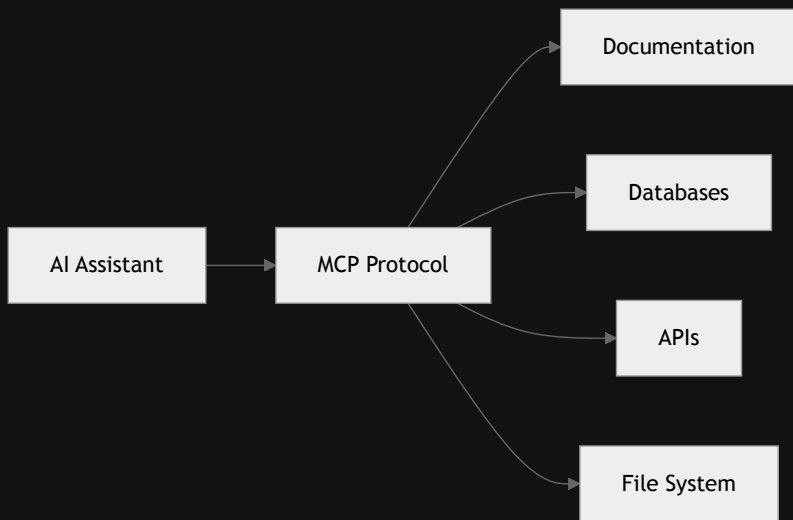
Model Context Protocol (MCP)

Connecting AI assistants to external tools and data sources

What is MCP?

Model Context Protocol - an open standard for AI-tool communication

- Universal interface for AI to access external capabilities
- Created by Anthropic, adopted across the industry
- Think of it as **"USB for AI"** - plug in any tool

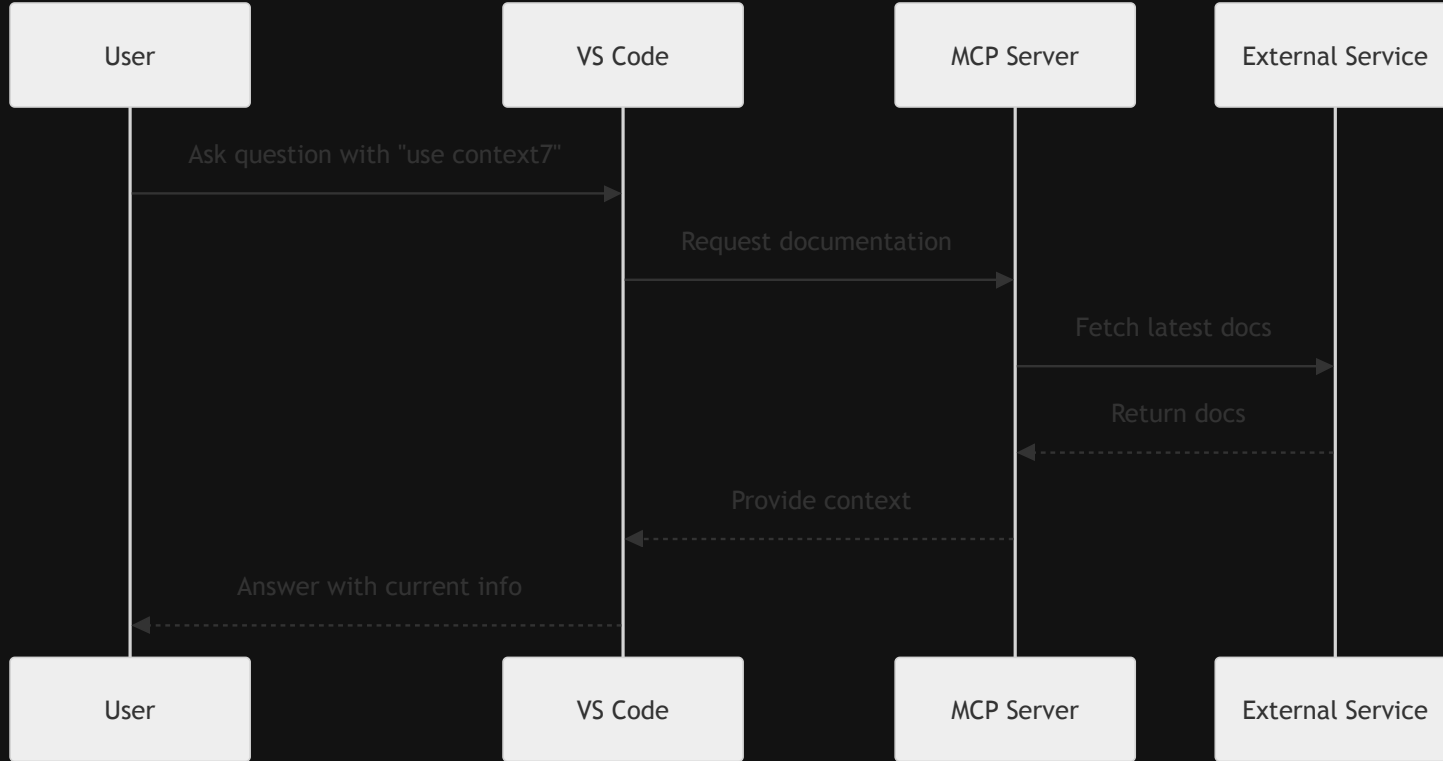


Why Do We Need MCP?

The Problem: LLMs have a knowledge cutoff date

Without MCP	With MCP
Outdated code examples	Up-to-date documentation
Hallucinated APIs	Real, verified APIs
Answers for old versions	Version-specific guidance
No access to your tools	Direct integration

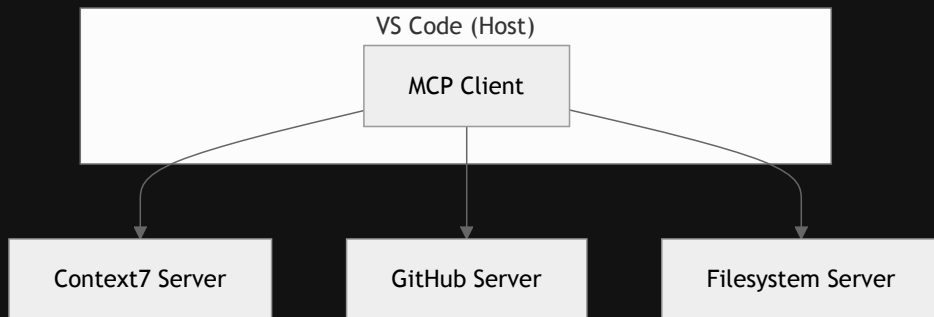
How MCP Works



MCP Architecture

Three key components:

- **Host:** The application (VS Code, Claude Desktop)
- **Client:** Protocol handler inside the host
- **Server:** Exposes tools, resources, and prompts



Examples of Popular MCP Servers

Server	Purpose
Context7	Up-to-date library documentation
GitHub	Issues, PRs, repository access
Filesystem	Read/write local files
PostgreSQL	Database queries
Puppeteer	Browser automation

Browse more: github.com/modelcontextprotocol/servers

What is Context7?

Context7 fetches current documentation directly into your prompts

Available tools:

- `resolve-library-id` - Find the Context7 ID for a library
- `get-library-docs` - Fetch documentation for a specific library

Supports 1000+ libraries including React, Next.js, Express, Prisma, and more

Setting Up Context7 in VS Code

Step 1: Create `.vscode/mcp.json` in your project

```
{
  "servers": {
    "context7": {
      "command": "npx",
      "args": ["-y", "@upstash/context7-mcp@latest"]
    }
  }
}
```

Starting the MCP Server

Step 2: Start the server in VS Code

1. Open Command Palette (`Cmd+Shift+P`)
2. Run "**MCP: List Servers**"
3. Select **context7**
4. Click "**Start Server**"

Or click the "**Start**" code lens in your `mcp.json` file

Verifying the Setup

Step 3: Confirm the server is running

- Open the Chat view (`Cmd+Shift+I`)
- Click the **Tools** icon in the chat box
- You should see Context7 tools listed:
 - `context7 → resolve-library-id`
 - `context7 → get-library-docs`

Using Context7 - Basic Usage

Add `use context7` **to your prompt:**

```
How do I set up authentication in an Express API? use context7 to fetch the relevant docs for the version I'm using.
```

Or specify a library directly:

```
Show me how I would create a prisma schema for my database. use library /prisma/prisma
```

Let's Try It!

Step 1: Set up Context7

1. Create `.vscode/mcp.json` with the configuration shown
2. Start the MCP server
3. Verify tools appear in Chat view

Step 2: Ask a documentation question

```
How do I create an Express router in my typescript project? use context7
```

Hands-On: More Examples

Try these prompts:

```
What's new in React 19? use context7
```

```
How do I set up Vitest with TypeScript in a Node project?  
use context7
```

```
How would I create a dockerfile for this app? Use context7
```

Pro Tip: Auto-Invoke Context7

Add a rule so you don't need to type "use context7" every time

Create `.github/copilot-instructions.md`:

```
Always use Context7 MCP when a request involves a library that the project uses.
```

Security Considerations

⚠ MCP servers can run arbitrary code on your machine

Best practices:

- Only install servers from **trusted sources**
- Review server configuration before starting
- VS Code prompts for **trust confirmation** on first start
- Check **server output logs** if something seems wrong

MCP: List Servers → Select server → Show Output

Security: What Servers Can Access

Capability	Risk Level	Example
Read files	Medium	Filesystem server
Write files	High	Could modify code
Network requests	Medium	API calls
Execute commands	VERY HIGH !!!	Shell access

Always review what permissions a server needs!

When NOT to Use MCP

✗ Don't use MCP when:

- Working with **proprietary/internal/confidential** client documentation
- Working in **air-gapped** environments
- **Without express approval from your project.**

✓ Use MCP when:

- You need **current** library documentation
- Working with **rapidly changing** elements of your project
- **You have permission from your project**

Troubleshooting

Server won't start?

1. Check Node.js version: `node --version` (need ≥ 18)
2. View server output: `MCP: List Servers` → `Show Output`
3. Reset and retry: `MCP: Reset Cached Tools`

Tools not appearing?

- Ensure server status shows "Running"
- Try restarting VS Code
- Check `mcp.json` syntax

Summary

- ✓ MCP provides a **standard protocol** for AI-tool integration
- ✓ MCP can give AI access to **up-to-date documentation**
- ✓ Setup is simple: **one JSON file** + start server
- ✓ Always consider **security** when adding MCP servers

Questions?

Resources:

- modelcontextprotocol.io - MCP Documentation
- context7.com - Context7 Website
- [VS Code MCP Docs](#)